# BUSINESS REPORT - PGP-DSGA

AUSTO MOTOR COMPANY
SAMIKSHA SINGH

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# Austo Automobile:

## Introduction:

Austo Motor Company, a leading producer of SUVs, Sedans, and Hatchbacks, recently convened its board to review the performance of its current marketing campaign. With concerns raised about its overall efficiency, the board has opted to bring in an analytics professional to strengthen and refine the campaign strategy.

## Data Description:

| Variable | Definition |
|---|---|
| Age | Age of the individual (years) |
| Gender | Gender (Male or Female or Unknown) |
| Profession | Occupation or profession |
| Marital_status | Marital status (Married & Single) |
| Education | Educational qualification as Graduate and Post Graduate |
| No_of_Dependents | The number of dependents (e.g., children, elderly parents) that the individual supports financially |
| Personal_loan | The individual has taken a personal loan "Yes" or "No" |
| House_loan | Whether the individual has taken a housing loan "Yes" or "No" |
| Partner_working | The individual's partner is employed "Yes" or "No" |
| Salary | The individual's salary or income |
| Partner_salary | The salary or income of the individual's partner |
| Total_salary | Combination of salary of the individual and their partner salary |
| Price | Price of a product or service |

| | |
|---|---|
| Make | Type of automobile such as SUV, Sedan, and Hatchback |

# Understanding the shape of the dataset:

There are 1581 rows and 14 columns

# Data Type

To determine the total salary, sum the individual's salary with the partner's salary using the formula: Total_Salary = Salary + Partner_Salary.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1581 entries, 0 to 1580
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Age               1581 non-null   int64
 1   Gender            1528 non-null   object
 2   Profession        1581 non-null   object
 3   Marital_status    1581 non-null   object
 4   Education         1581 non-null   object
 5   No_of_Dependents  1581 non-null   int64
 6   Personal_loan     1581 non-null   object
 7   House_loan        1581 non-null   object
 8   Partner_working   1581 non-null   object
 9   Salary            1581 non-null   int64
 10  Partner_salary    1475 non-null   float64
 11  Total_salary      1581 non-null   int64
 12  Price             1581 non-null   int64
 13  Make              1581 non-null   object
dtypes: float64(1), int64(5), object(8)
memory usage: 173.1+ KB
```

*Table 1: dataset details*

- The dataset contains **1,581 entries** and **14 columns**, all mostly complete.
- Only one column, **Partner_salary**, has missing values (106 entries).

- The **Total_salary** column is already computed, likely as the sum of **Salary** and **Partner_salary**.
- Data types include **int64**, **float64**, and **object**, suitable for both statistical and categorical analysis.
- Features like **Gender**, **Age**, **Profession**, and **Marital_status** offer strong potential for demographic insights.

## Getting the statistical summary for the dataset:

| | Age | No_of_Dependents | Salary | Partner_salary | Total_salary | Price |
|---|---|---|---|---|---|---|
| **count** | 1581.000000 | 1581.000000 | 1581.000000 | 1475.000000 | 1581.000000 | 1581.000000 |
| **mean** | 31.922201 | 2.457938 | 60392.220114 | 20225.559322 | 79625.996205 | 35597.722960 |
| **std** | 8.425978 | 0.943483 | 14674.825044 | 19573.149277 | 25545.857768 | 13633.636545 |
| **min** | 22.000000 | 0.000000 | 30000.000000 | 0.000000 | 30000.000000 | 18000.000000 |
| **25%** | 25.000000 | 2.000000 | 51900.000000 | 0.000000 | 60500.000000 | 25000.000000 |
| **50%** | 29.000000 | 2.000000 | 59500.000000 | 25600.000000 | 78000.000000 | 31000.000000 |
| **75%** | 38.000000 | 3.000000 | 71800.000000 | 38300.000000 | 95900.000000 | 47000.000000 |
| **max** | 54.000000 | 4.000000 | 99300.000000 | 80500.000000 | 171000.000000 | 70000.000000 |

*Table 2: Statistical data details*

The dataset comprises **1,581 individuals**, with all columns fully populated except for **Partner_salary**, which has **106 missing entries**.

- The average age is **31.9 years**, with the maximum being **54 years**.
- The average number of dependents is **2.45**, indicating most individuals have **2–3 dependents**.
- The **average salary** is approximately **₹60,000**, while the **average partner salary** is around **₹19,000**.
- The **combined average household income** is roughly **₹80,000**, with the highest reaching **₹1,71,000**.

- The **average vehicle purchase price** is about **₹36,000**, and the **maximum price** is **₹70,000**
- The dataset contains missing values in the **Gender** and **Partner_salary** columns, with **53** and **106** null entries respectively. No duplicate records were identified.
- Missing values in **Partner_salary** were addressed through appropriate imputation techniques. Spelling inconsistencies in the **Gender** column were corrected, and missing entries were filled with the label **'Unknown'**

# UNIVARIATE ANALYSIS FOR NUMERICAL VARIABLE DATA:

## 1. No_of_Dependent:

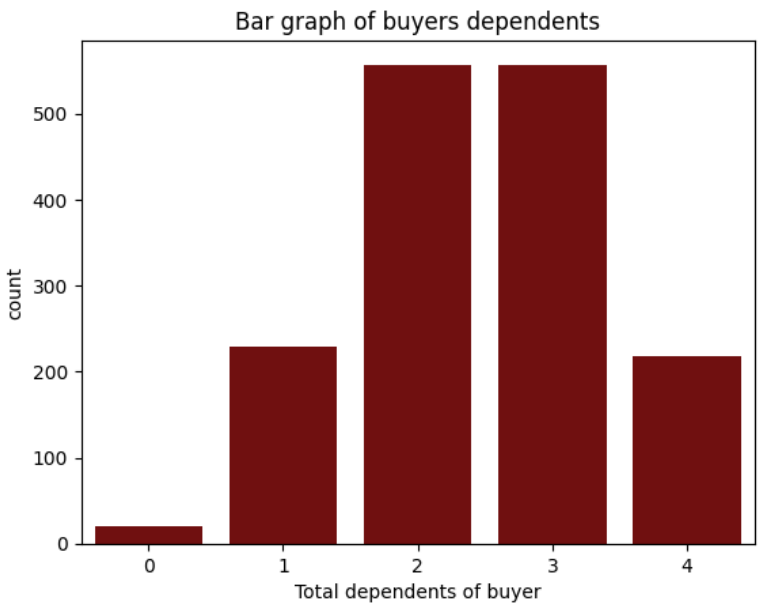
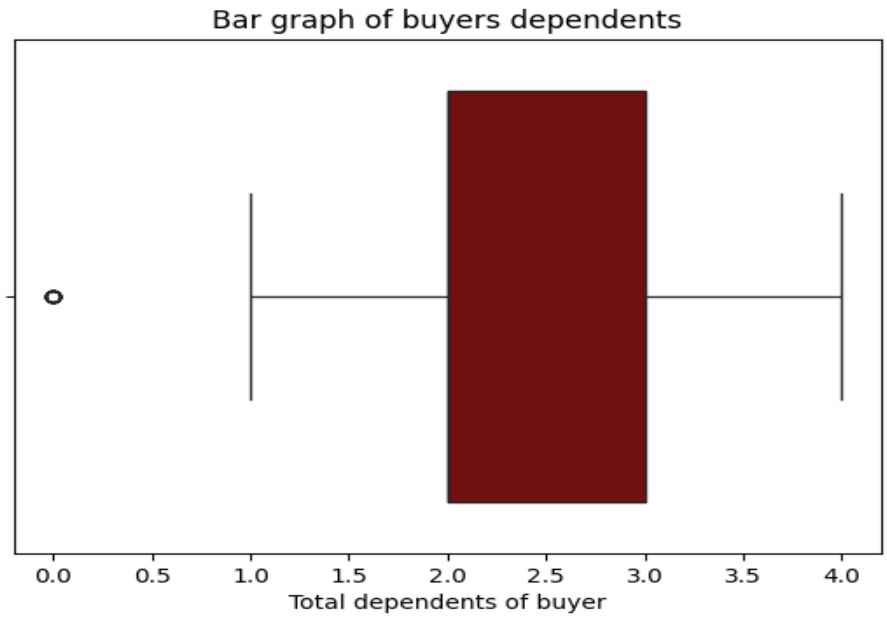



*Figure A. 1.1 Histplot for No_of_Dependents*




*Boxplot for Age variable*

- Most buyers have **2 or 3 dependents**, making it the most common family size in the dataset.
- Buyers with **0 dependents** are rare and appear as outliers.
- The **median number of dependents** is around **2.5**, based on the box plot.

## Age:

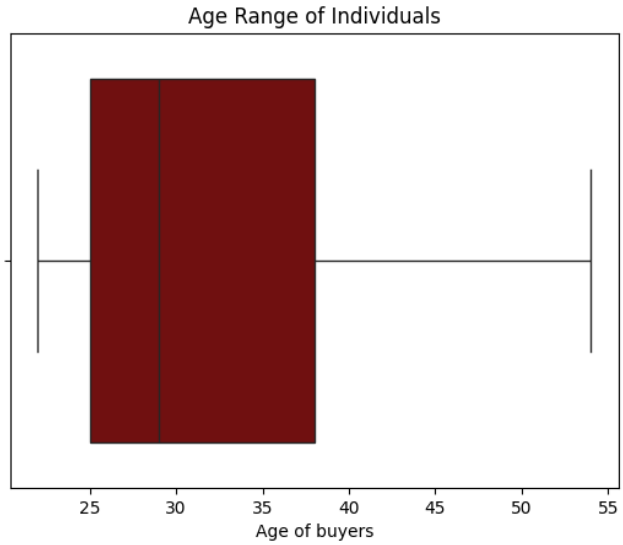
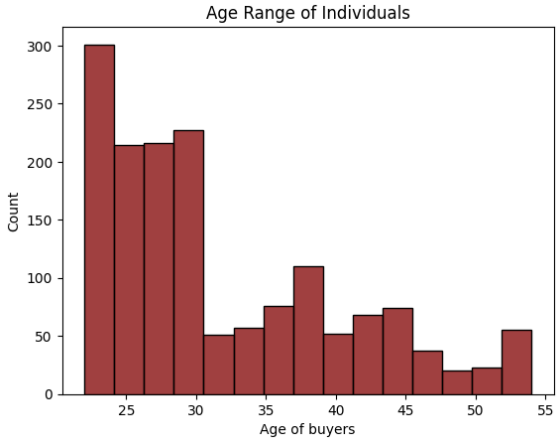



*Figure A.1.2 Histplot for Age*




*Boxplot for Age variable*

- The customer base skews young, with the 20-25 age bracket representing the largest segment.
- The distribution centers around a median age of 30, encompassing a core demographic of 25- to 40-year-olds.
- While the total age range extends to 55, representation diminishes considerably among older cohorts.

## Salary:

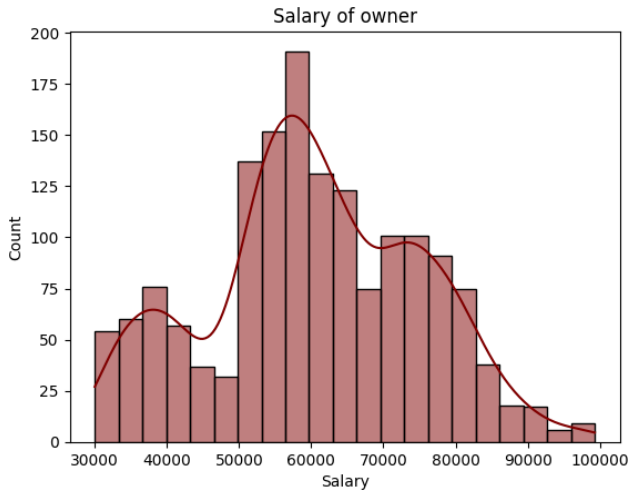
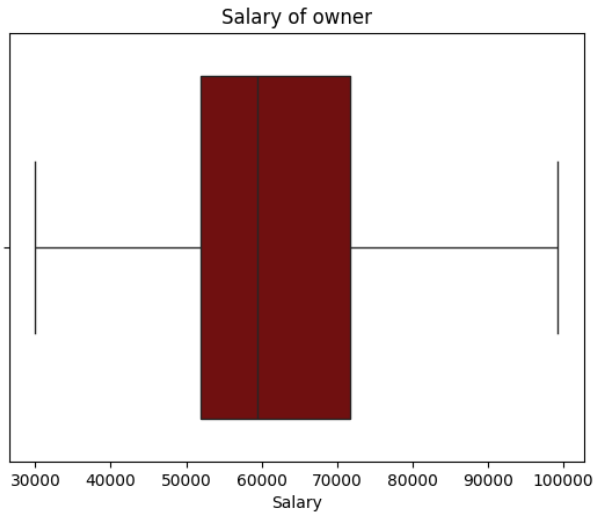



*Figure A.1.3 Histplot for Salary variable*




*Boxplot for Salary Variable*

- Owner salaries peak in the ₹55,000–₹60,000 range, representing the most common earning bracket.

- The median salary stands at approximately ₹60,000, with the bulk of salaries concentrated between ₹50,000 and ₹70,000.

- The salary range spans from ₹30,000 to ₹99,300, reflecting significant income variation across the dataset

## Partner Salary:

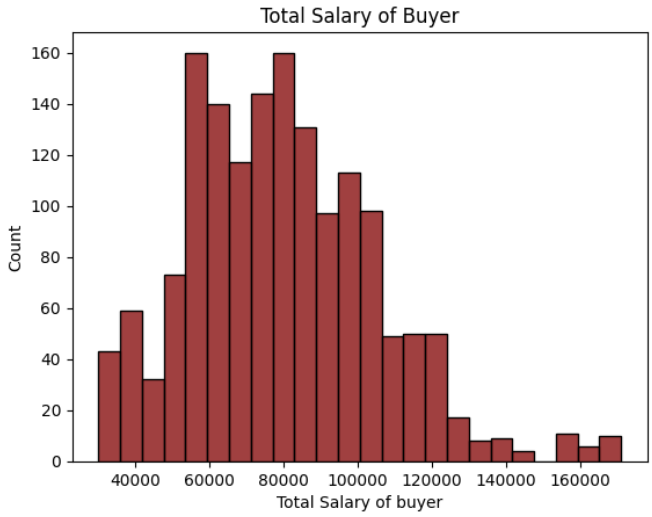
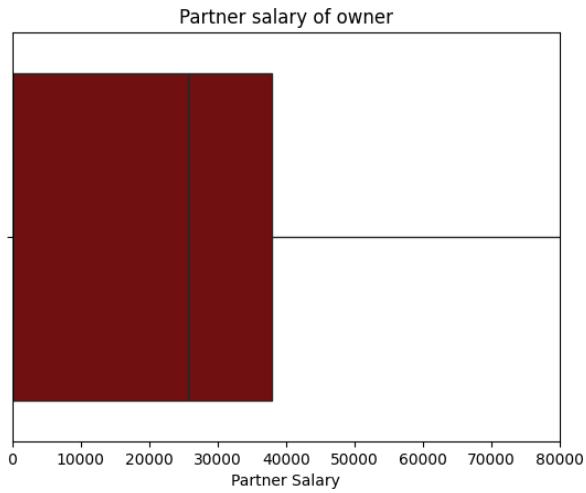



*Figure A.1.4 Histplot for Partner_salary Variable*




*Boxplot for Partner_salary variable*

- A substantial share of owners report partners with ₹0 earnings, which constitutes the predominant salary bracket among partners.
- The median partner salary reaches approximately ₹30,000, with the majority clustered between ₹10,000 and ₹40,000.
- While partner salaries extend to ₹80,500, higher income brackets represent a significantly smaller portion of the distribution.

## Total Salary:

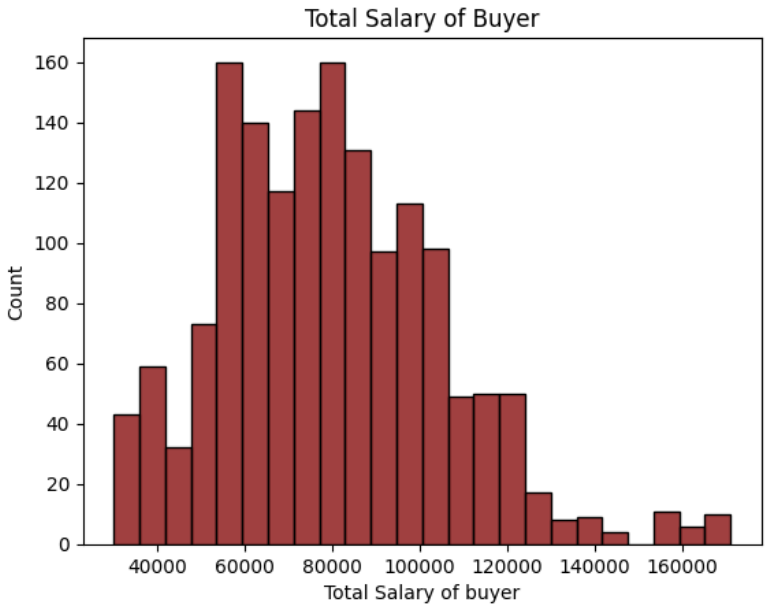
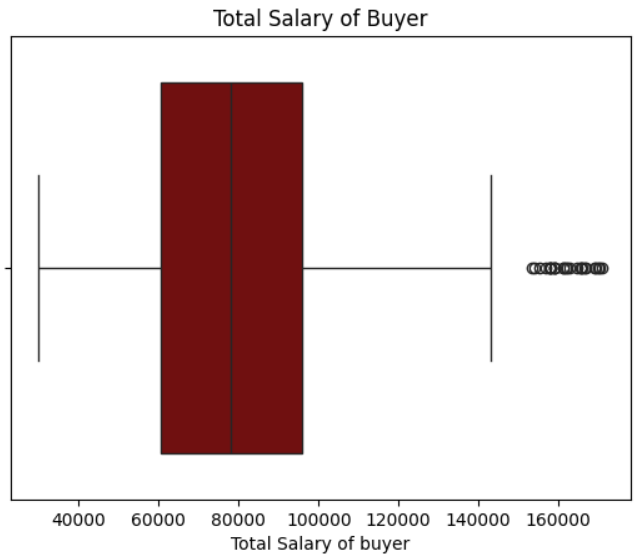



*Figure A.1.5 Histplot for Total_salary variable*




*Boxplot for Total_salary variable*

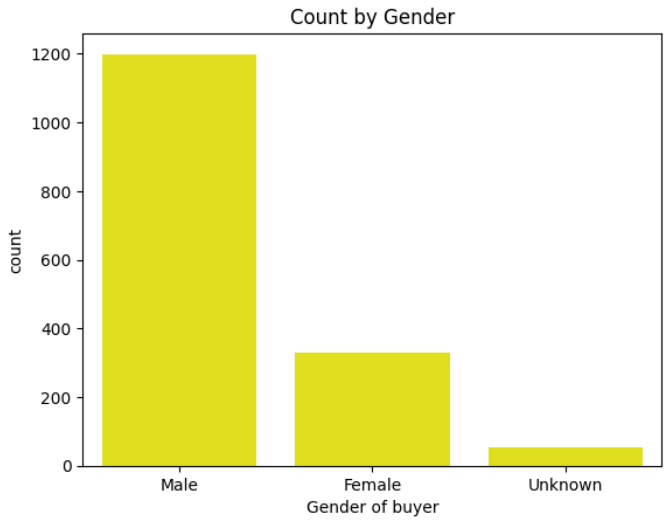
- The majority of buyers fall within the ₹50,000 to ₹100,000 earnings band, with peak concentration occurring between ₹60,000 and ₹70,000.

- The median total salary is approximately ₹80,000, with most earners positioned between ₹60,000 and ₹100,000.

- Salaries extend from ₹30,000 at the lower threshold to ₹171,000 at the upper end, with scattered high-income outliers exceeding ₹140,000.

## UNIVARIATE ANALYSIS FOR CATEGORICAL VARIABLE DATA:

### Gender:




| Gender | |
|---|---|
| Male | 1199 |
| Female | 329 |
| Unknown | 53 |

*Figure A.2.1 Histplot for Gender variable*

• The data for gender states that Male has more interest in buying a car over female.

• The unknown value denotes that the specific data is not available as it could be either male or female.

## Profession:

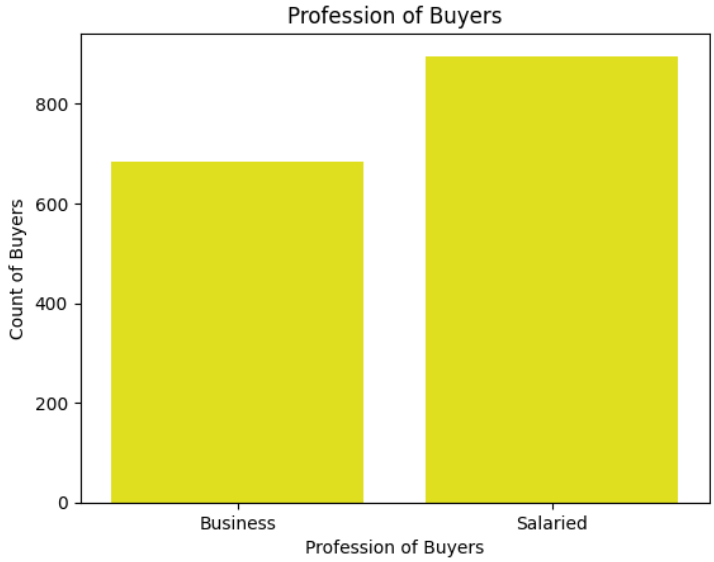



*Figure A.2.2 Histplot for Profession variable*

## Marital Status:

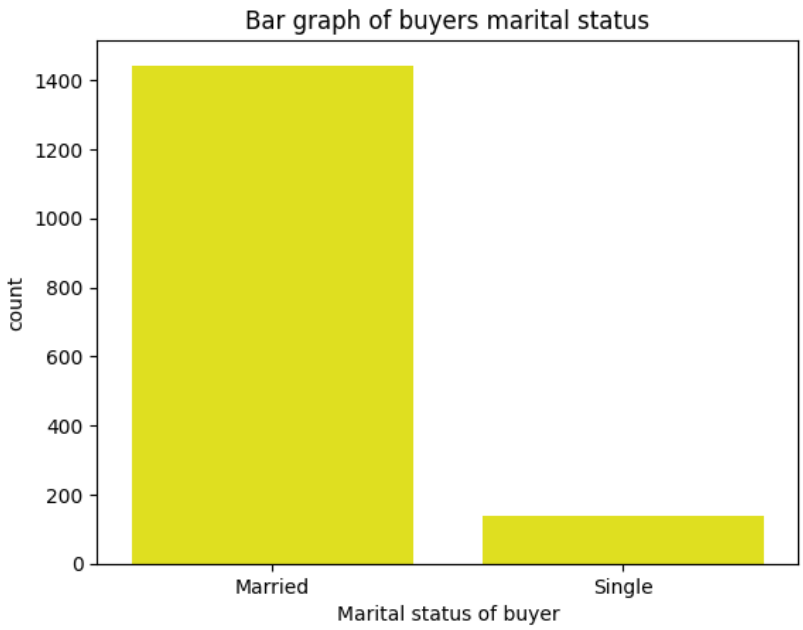



*Figure A.2.3 Histplot for Martial_status variable*

## Education:

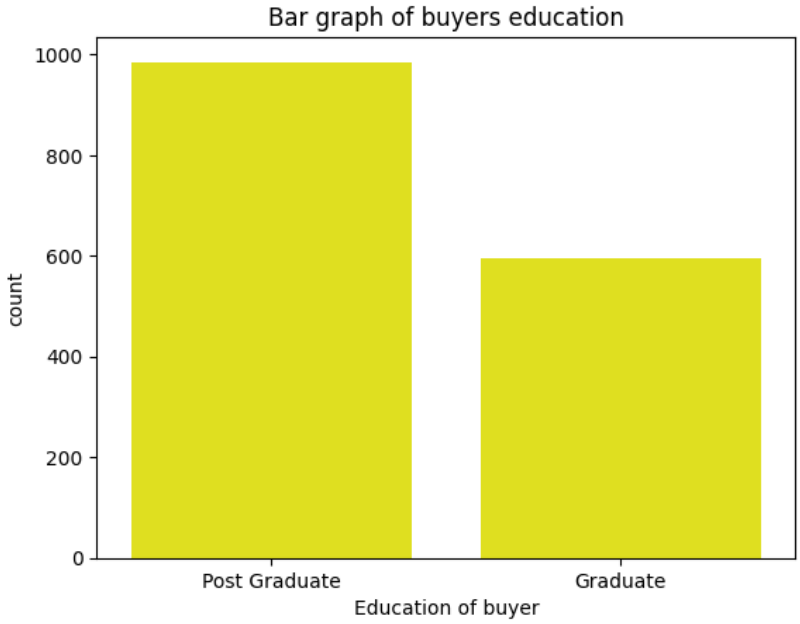



*Figure A.2.4 Histplot for Education variable*

## Partner Working:

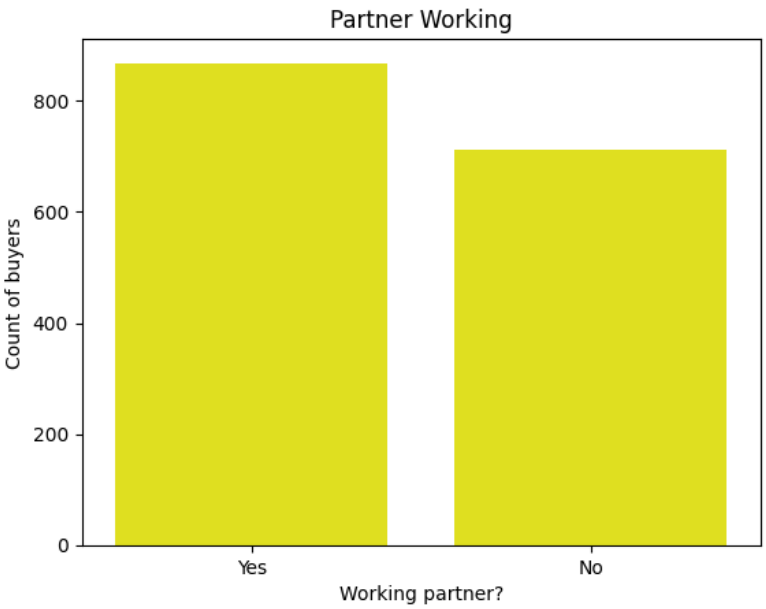



*Figure A.2.5 Histplot for Partner_working variable*

## Make:

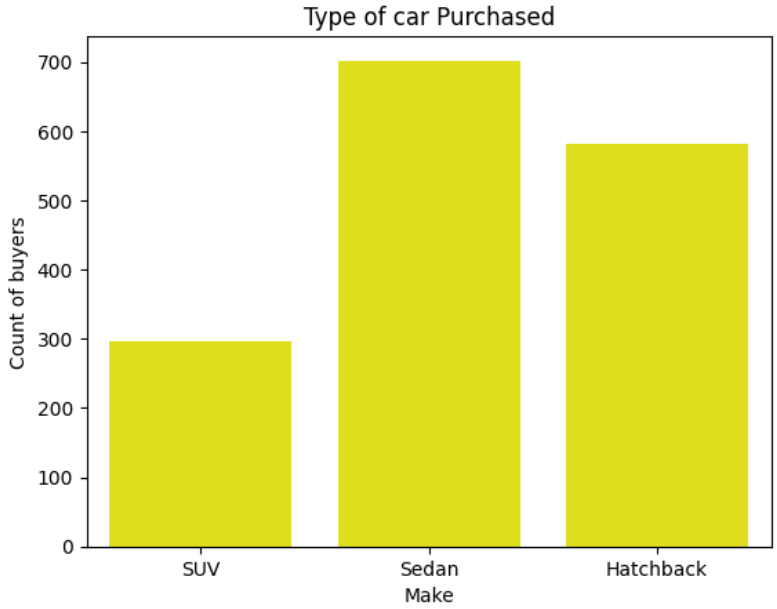



*Figure A.2.6 Histplot for Make variable*

## Personal Loan:

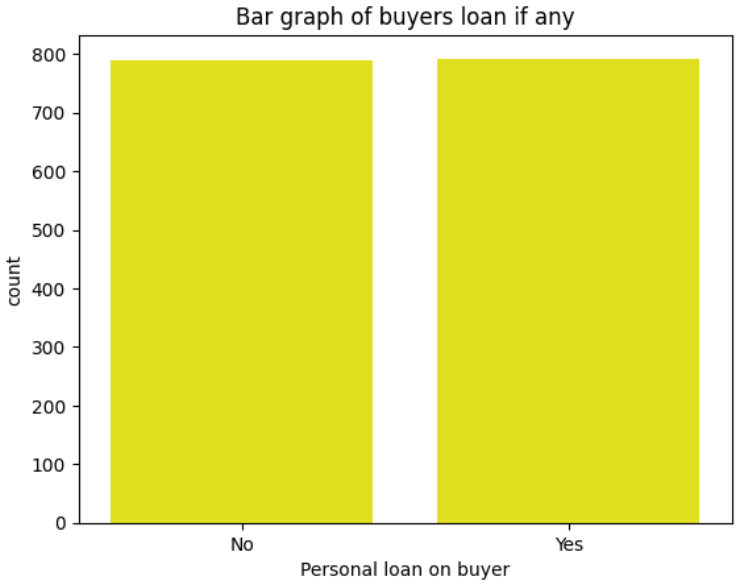



*Figure A.2.7 Histplot for Personal_loan variable*

## House Loan

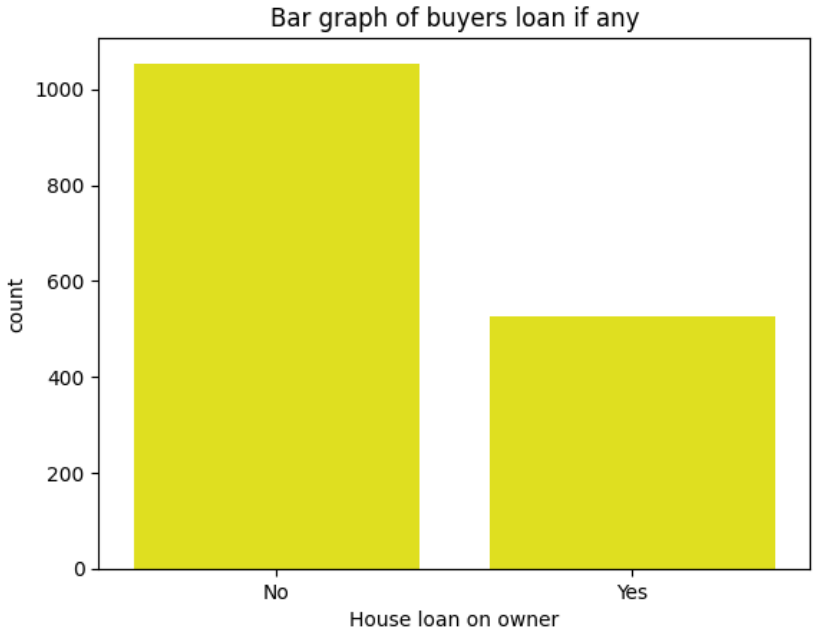



*Figure A.2.8 Histplot for House_loan variable*

## OBSERVATION AND INSIGHTS:

## INSIGHTS:

- Most buyers **don't have a house loan**; fewer than half do.
- **Personal loans** are evenly split—about half have one, half don't.
- **Sedans** are the most popular car type, followed by hatchbacks, then SUVs.
- More buyers have **working partners** than not.
- Most buyers are **postgraduates**; fewer are graduates.
- The majority of buyers are **married**.
- More buyers are **salaried** than in business.

## Observations

The **price** of vehicles shows a positive skew, suggesting that the majority of buyers prefer **mid-range cars**, while only a small number opt for **high-end, expensive vehicles**. Overall, most customers belong to the **salaried class**, reflecting stable income levels that influence their purchasing decisions.

# Bivariate Analysis

## 1. Numerical vs Numerical

- Exploring the relationship between all numerical variables
-

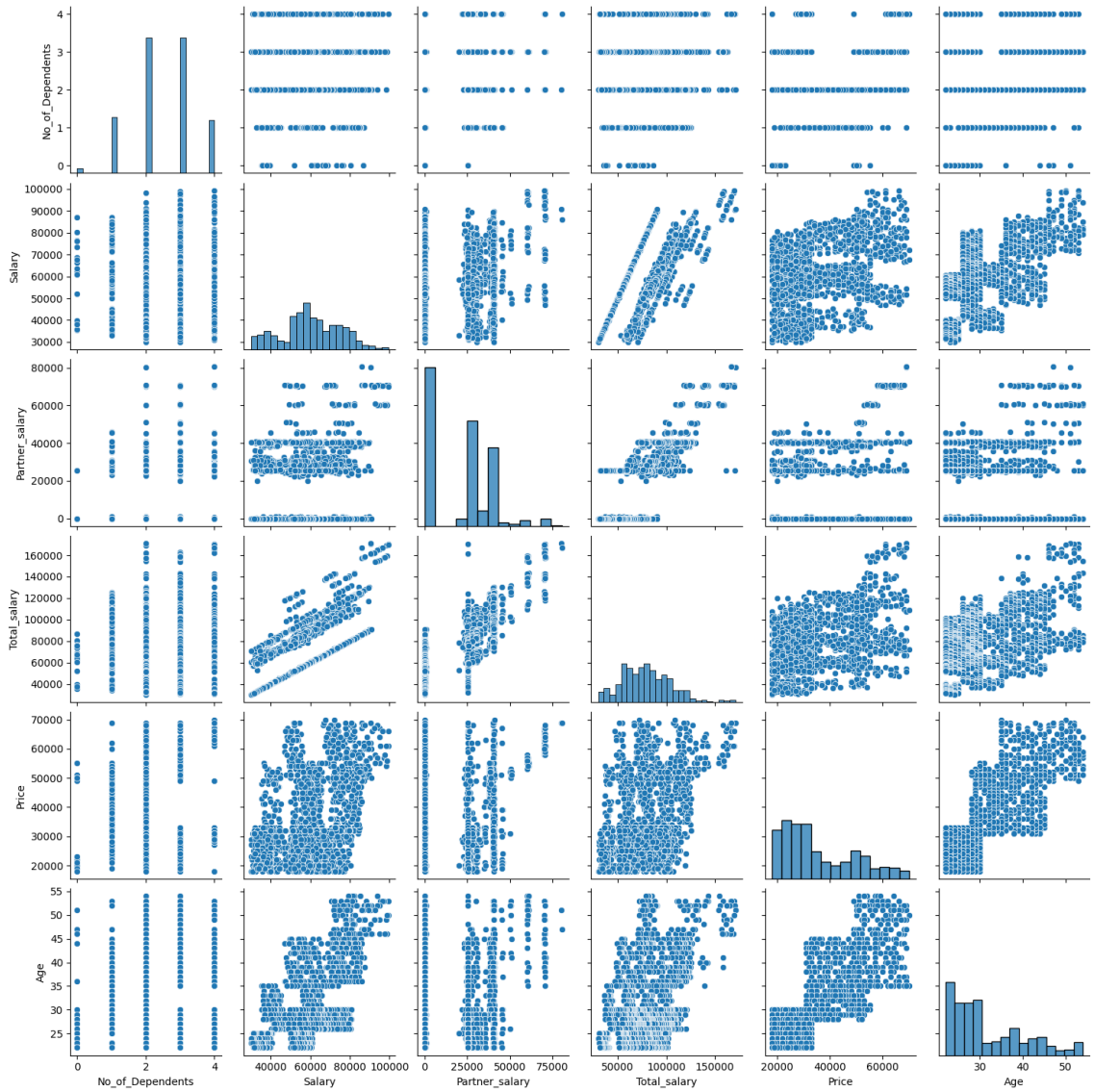



*Figure B.1.1 Numerical data correlation*

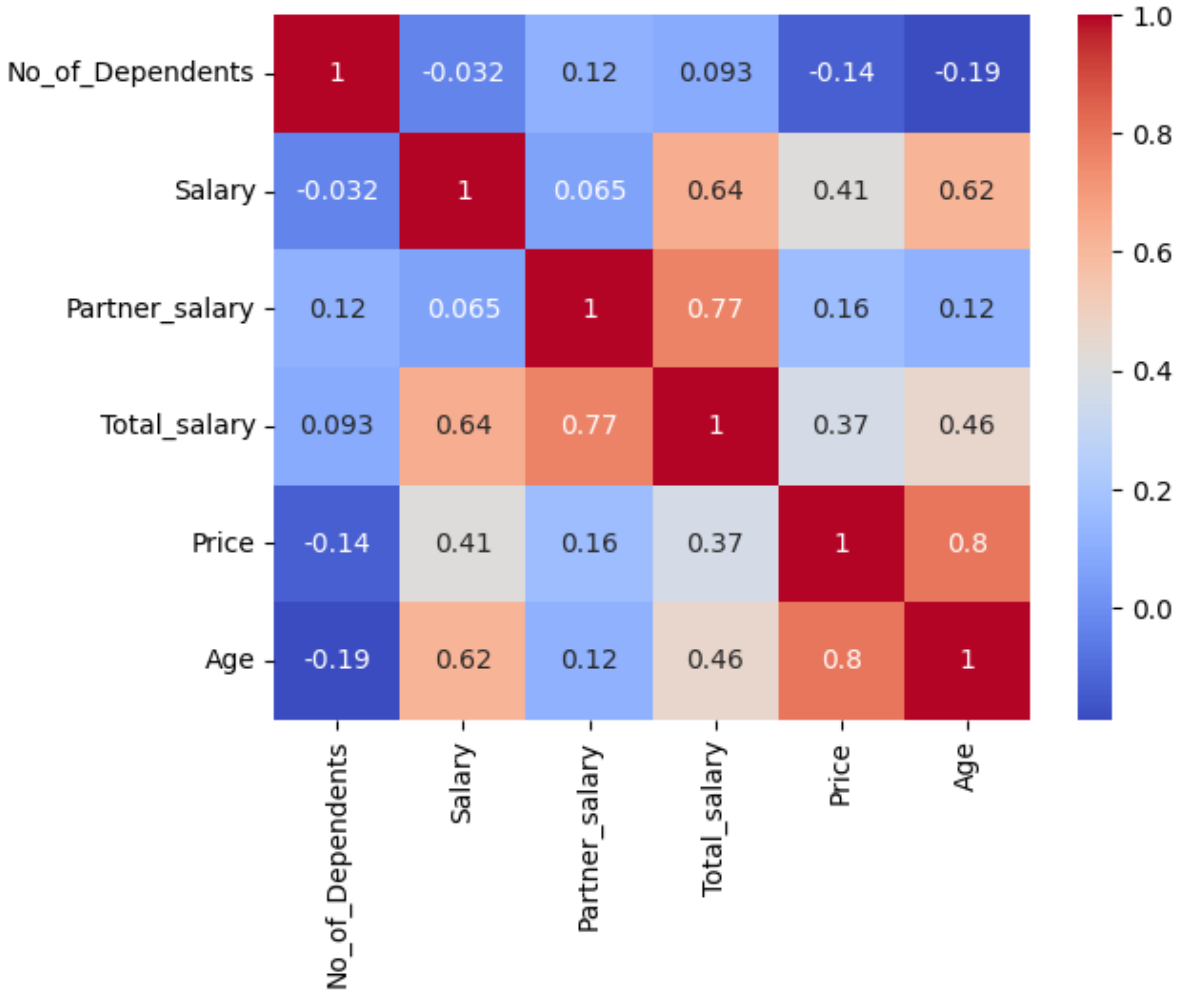



*Figure B.1.2 Numerical data correlation*

## Positive Correlations

- **Car Price & Buyer Age:** There's a strong positive correlation (0.81) between the price of cars and the age of buyers, indicating that older individuals tend to purchase more expensive vehicles.
- **Total Salary & Partner's Salary:** These two variables also show a positive correlation, meaning that as one increases, the other tends to rise as well.

**Negative Correlations**

- **Number of Dependents vs Salary & Car Price:** A negative correlation exists between the number of dependents and both the buyer's salary and the price of cars. This suggests that as the number of dependents increases, total income and spending on cars tend to decrease.

## 2. Numerical vs Categorical

- Exploring the relationship between categorical vs numerical variables

**Price Vs Gender:**

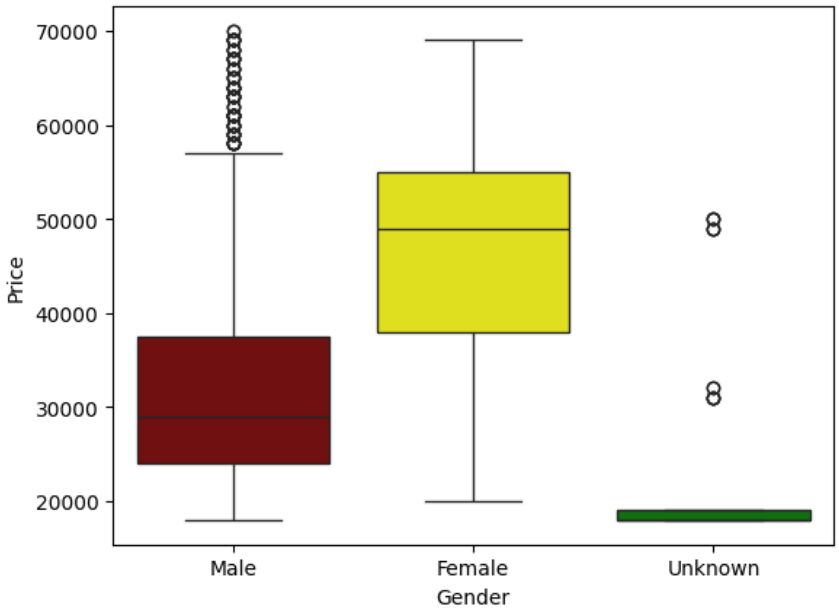



*Figure B.2.3.1 One Numerical and one Categorical graph*

**Female buyers** tend to purchase cars at **higher median prices** than male buyers.

- **Male buyers** show a wider spread in prices, with more **high-end outliers**.

- Buyers labelled as **'Unknown' gender** have the **lowest median price** and a very narrow price range.

## Age Vs Make

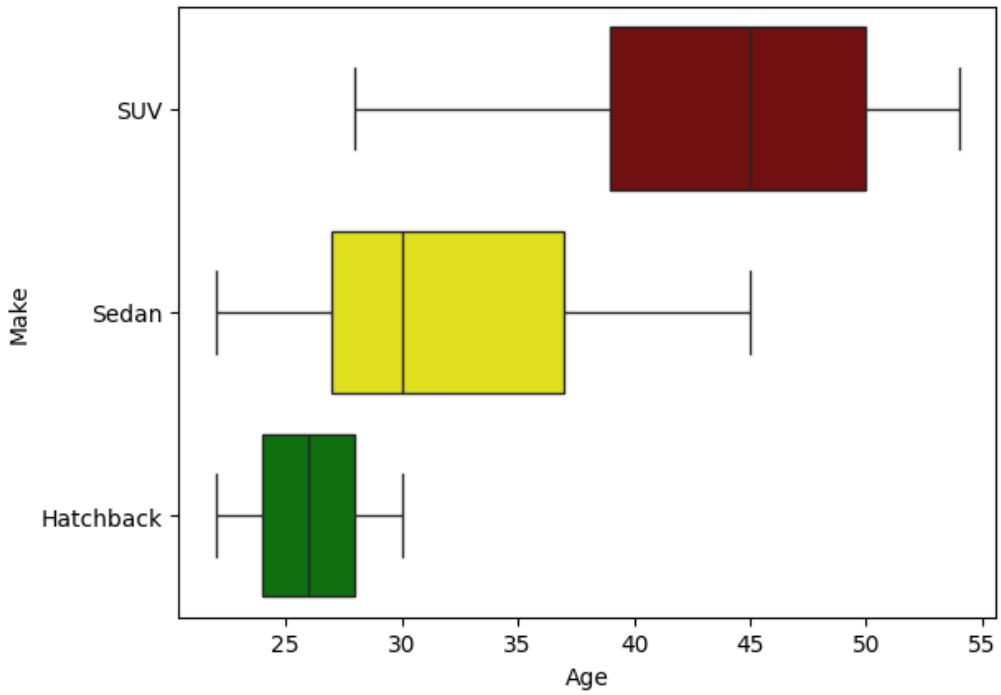



*Figure B.2.3.2 One Numerical and one Categorical graph*

- SUV buyers tend to be older, with most around 45 years old.
- Sedan buyers are typically in their mid-30s.
- Hatchback buyers are the youngest group, mostly around 28 years old

## Age Vs  Price

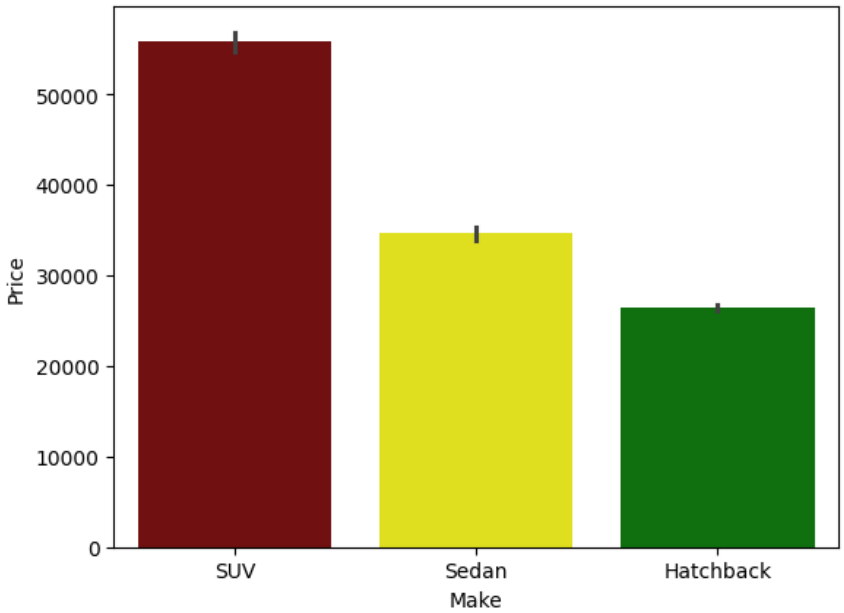



*Figure B.2.3.3 One Numerical and one Categorical graph*

- **SUVs** dominate in price, averaging just above ₹60,000, suggesting a premium market segment.
- **Sedans** sit in the middle range (~₹40,000), likely appealing to mid-tier consumers.
- **Hatchbacks** are the most budget-friendly (~₹30,000), ideal for cost-conscious buyers or urban commuters.
- The **error bars** hint at variability—possibly due to brand differences or optional features.

**Salary Vs Gender**

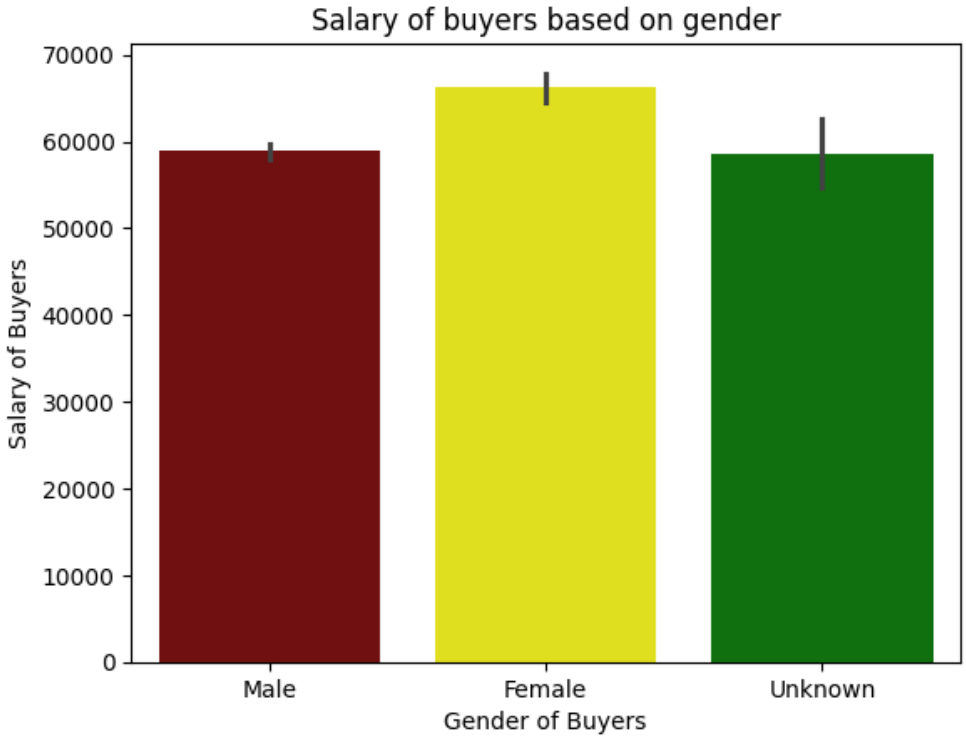



*Figure B.2.3.4 One Numerical and one Categorical graph*

- **Female buyers** show the highest average salary, slightly above ₹60,000, which challenges some common assumptions and could be a powerful point in your report.
- **Male and Unknown gender** categories are nearly tied, just below ₹60,000, suggesting similar purchasing power.
- The **error bars** again indicate variability—possibly due to sample size or outliers in each group.

## No. of Dependents Vs Price

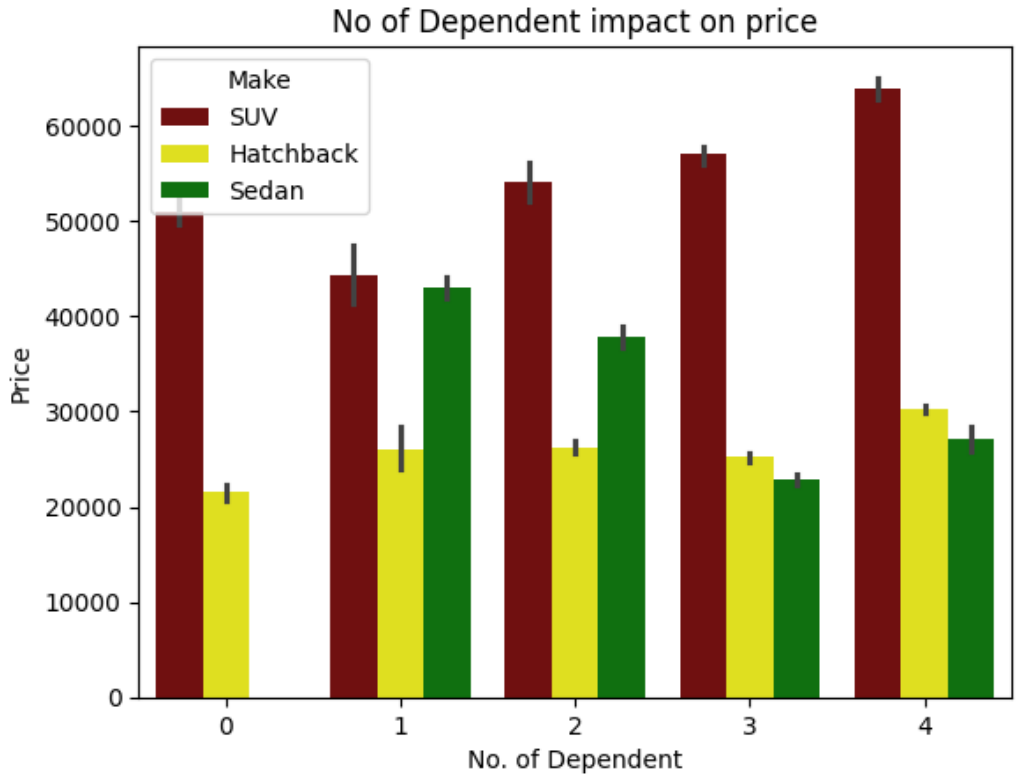



*Figure B.2.3.5 One Numerical and one Categorical graph*

- **Female buyers** show the highest average salary, slightly above ₹60,000, which challenges some common assumptions and could be a powerful point in your report.
- **Male and Unknown gender** categories are nearly tied, just below ₹60,000, suggesting similar purchasing power.

## Salary Vs Make

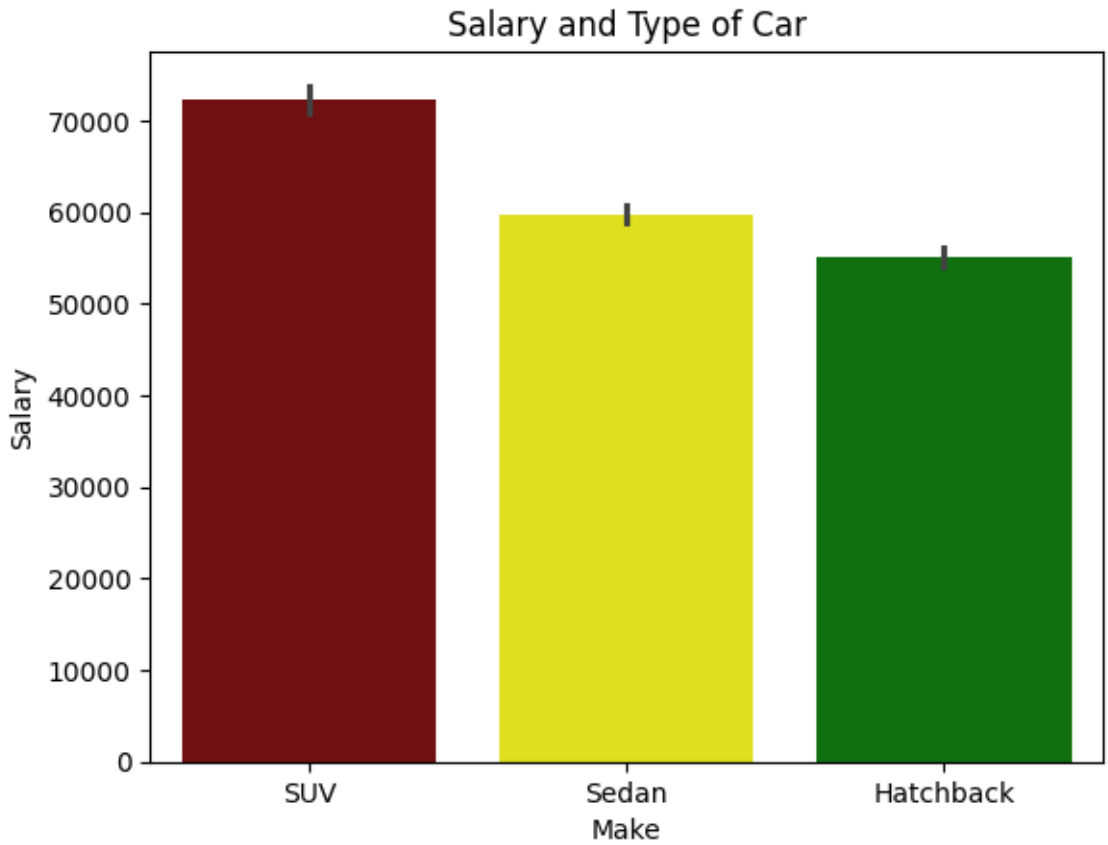



*Figure B.2.3.6 One Numerical and one Categorical graph*

- **Income directly dictates vehicle choice**, creating a clear hierarchy from premium SUVs down to practical hatchbacks.
- **Sedans occupy the strategic middle ground**, catering to those who seek status without the top-tier price tag.
- **Hatchbacks define the entry point**, serving as the primary vehicle for consumers with the most budget constraints.

## KEY QUESTIONS:

**1.** Do men tend to prefer SUVs more compared to women?

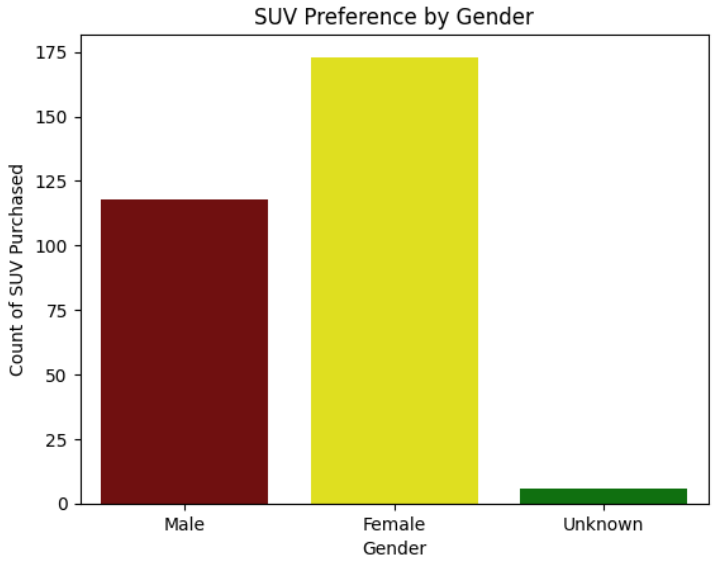



*Figure 1. Comparison graph of Gender and Make*

**Women** tend to show a **stronger** preference for purchasing **SUVs** compared to men, who are more inclined toward hatchbacks and sedans.

**2.** What is the likelihood of a salaried person buying a Sedan?

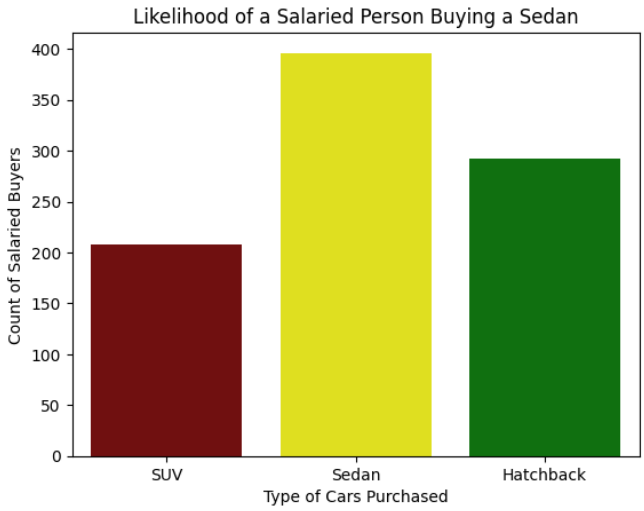



*Figure 2. Comparison graph of Salary and Make*

It can be concluded that salaried individuals are more likely to choose sedans over other car types.

**3.** What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

| Gender | Profession | Hatchback | SUV | Sedan |
|---|---|---|---|---|
| Female | Business | 0.000000 | 52.380952 | 47.619048 |
| | Salaried | 6.696429 | 52.678571 | 40.625000 |
| Male | Business | 51.699463 | 5.903399 | 42.397138 |
| | Salaried | 43.125000 | 13.281250 | 43.593750 |
| Unknown | Business | 4.761905 | 4.761905 | 90.476190 |
| | Salaried | 3.125000 | 15.625000 | 81.250000 |

*Figure 3. Descriptive statistics table about Profession and Product model*

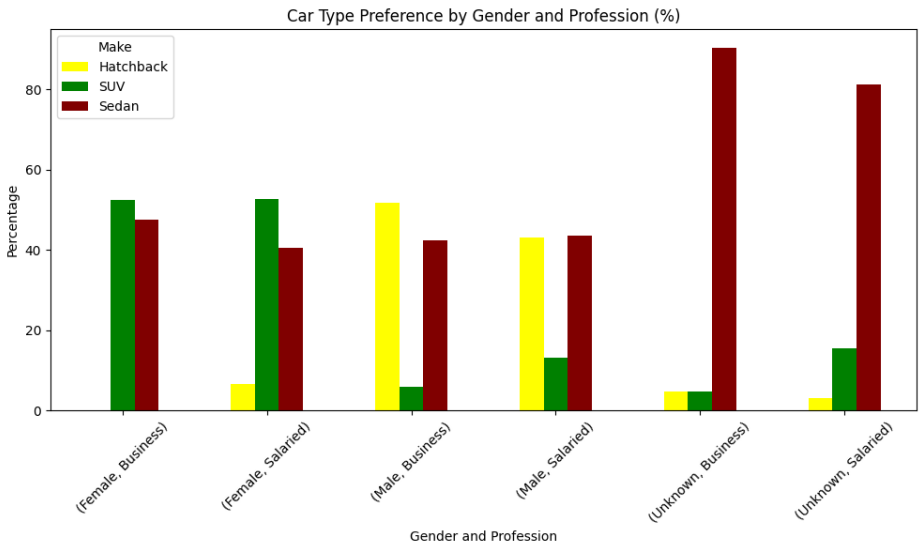



*Figure 4. Comparison graph about Profession and Gender*

When it comes to car choices, salaried males seem to pick sedans over SUVs

**4.** How does the amount spent on purchasing automobiles vary by gender?

| Gender | Price |
|---|---|
| **Female** | 15695000 |
| **Male** | 39348000 |
| **Unknown** | 1237000 |

*Figure 5. Statistics table on Gender and Price*

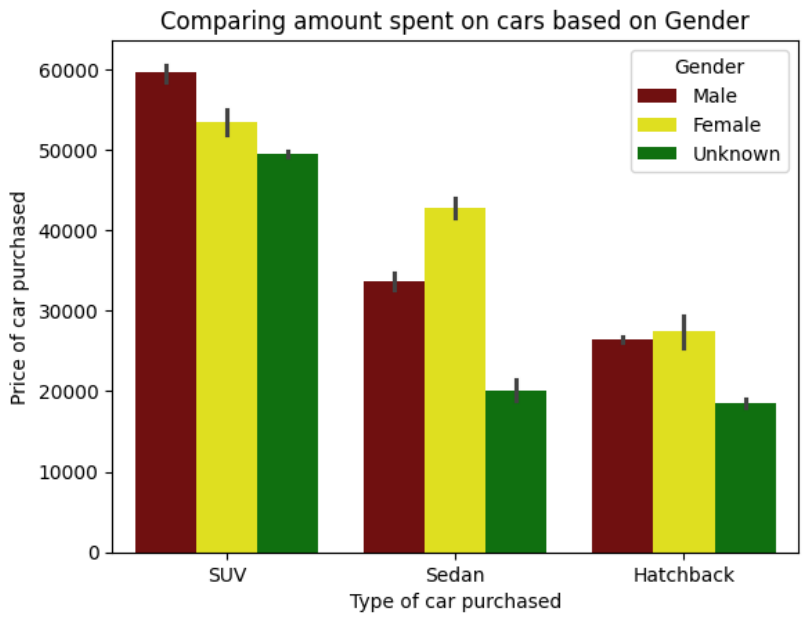



*Figure 6. Comparison graph about Price and Make*

- Males spend slightly more on SUVs than females, though both genders show strong interest in this category.

- Females lead in sedan purchases, spending significantly more than males and others.
- Hatchback spending is nearly equal between males and females, with only a minor difference.
- Overall, males account for the highest total car expenditure, followed by females, with the 'Unknown' category trailing far behind.

**5.** How much money was spent on purchasing automobiles by individuals who took a personal loan?

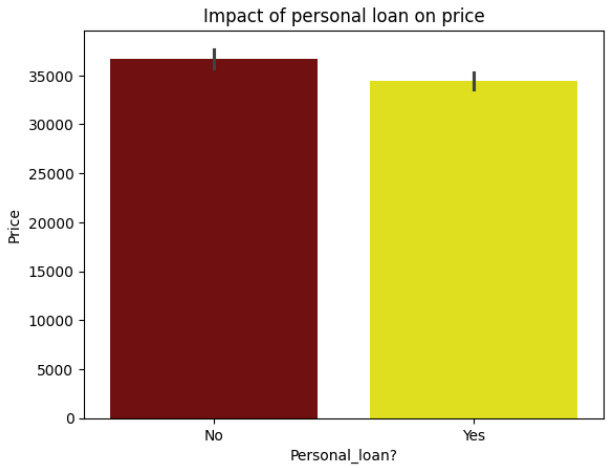



*Figure 7. Comparison graph about Price and Personal_loan*

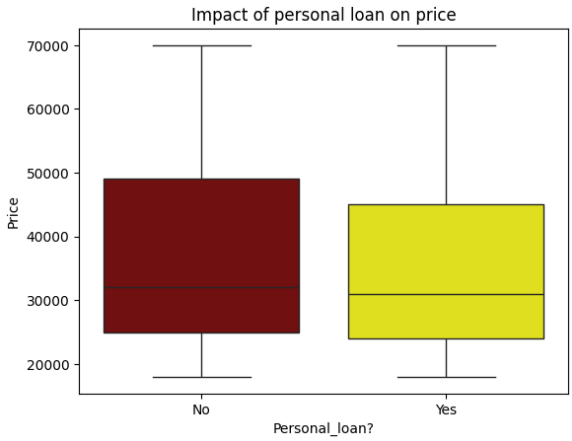



*Figure 7. Comparison graph about Price and Personal_loan*

- People who didn’t take a personal loan actually spent a bit more on their cars than those who did. It’s a small difference, but it’s there.
- On average, non-loan buyers crossed the ₹37,000 mark, while loan-takers stayed just below it.
- The median price is pretty similar for both groups, so it’s not like loan-takers are buying drastically cheaper cars.
- What’s interesting is the spread—those with loans tend to stick to a tighter budget, while non-loan buyers show more variation in how much they spend.
- So overall, taking a personal loan doesn’t seem to push people toward bigger purchases. If anything, it might make them a bit more cautious.

**6.** How does having a working partner influence the purchase of higher-priced cars?

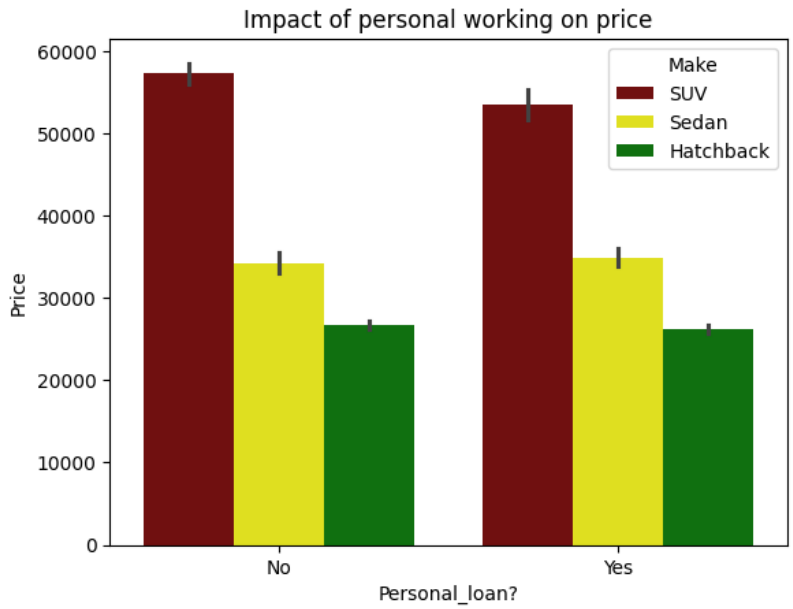



*Figure 8. Comparison graph about Price and Personal_loan*

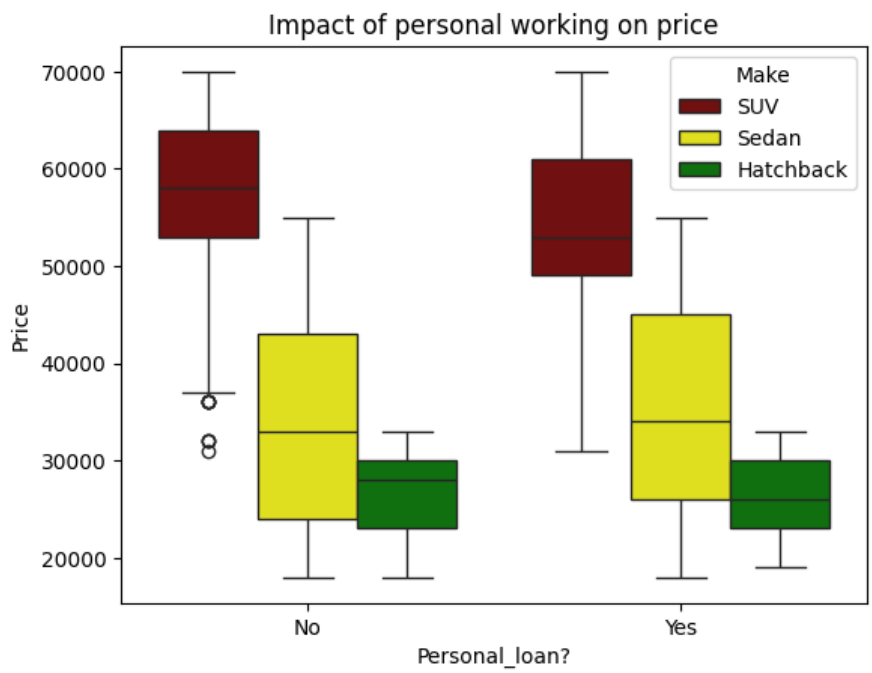



*Figure 9. Comparison graph about Price and Personal_loan*

- A non-working partner may lead to tighter household budgets, reducing the likelihood of buying higher-priced cars

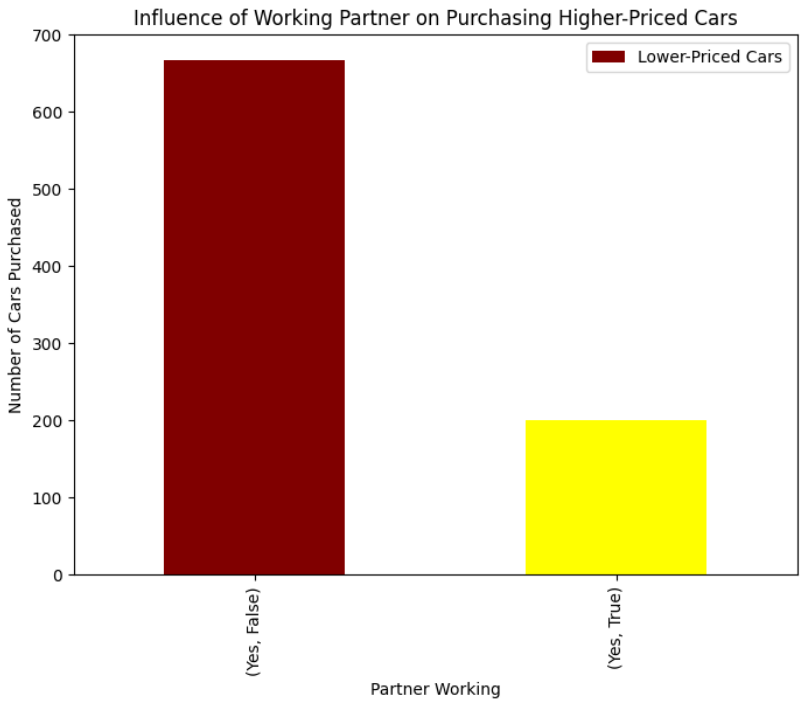



*Figure 10. Comparison graph about Number of Cars Purchased and Partner_working*

- **Dual-income households** represent a prime target for **higher-end vehicles** like SUVs and premium Sedans.
- Marketing campaigns should emphasize **comfort, status, and long-term value** to appeal to these buyers.

# Actionable Insights & Recommendations

## Actionable Insights

1. **Diverse Customer Segments**
   Austo Motors serves distinct customer groups — families, professionals, and young buyers — each with different expectations and purchase motivations. A one-size-fits-all approach limits engagement and sales potential.
2. **SUV Dominance and Loan Dependence**
   SUVs continue to be the most popular choice, especially among family-oriented buyers. Many of these customers rely on personal loans, indicating that financing options strongly influence purchasing decisions.
3. **Price Sensitivity in Hatchbacks**
   Hatchback buyers are highly price-conscious and value fuel efficiency, affordability, and convenience over luxury or size.
4. **Localized Inventory Needs**
   Sales success varies by region and demographic. Areas with younger populations perform better with hatchbacks, while suburban and family-oriented regions show higher SUV demand.
5. **Marketing Mismatch**
   Current marketing campaigns are too general and fail to address the unique motivations of each customer segment

## Business Recommendations

1. **Refine Marketing Messaging**

**SUVs:** Focus campaigns on family, safety, comfort, and reliability. Use media platforms that reach older and family-centric audiences.

**Sedans:** Highlight elegance, technology, and value for professionals. Use digital media, career platforms, and lifestyle magazines.

**Hatchbacks:** Emphasize economy, fun, and convenience. Use social media and youth-focused marketing.

2. **Develop Strategic Partnerships**

Partner with banks and financial institutions to offer pre-approved or low-interest car loans, especially targeting SUV buyers.

Promote these financing options in campaigns to make purchasing smoother and more appealing.

3. **Optimize Sales and Inventory**

Train sales teams to quickly identify a buyer's profile and tailor recommendations accordingly.

Adjust dealership inventory based on local demographics:

More hatchbacks in areas with younger populations.

More SUVs in family-heavy suburbs.

4. **Guide Product Development**

Expand the SUV lineup with varied price points and model sizes to capture growing demand.

Keep hatchbacks competitively priced with strong mileage and low maintenance costs to maintain their market appeal.